
EECS 16A Designing Information Devices and Systems I

Fall 2023 Pre-Lab Reading: Imaging 1

1 Introduction

Pre-Lab readings are short 1-2 page notes that we will have you read before your lab sections, and future homeworks will have a few questions related to them. They will contain a brief recap of previous lab material as well as a short overview of what to expect for the upcoming lab(s) of the week.

2 Logistics/Announcements

- (a) We will be starting the **Imaging Module** on Monday, September 11. Starting from Monday's lab (Imaging 1), all labs will be **graded** (i.e. graded checkoff after completing the lab).
- (b) **Important:** We will be giving you your lab kits during your respective lab sections starting Monday, September 11. Although many of you may be working with a partner, every student should receive their own lab kit. Specific instructions on how to collect your lab kits will be given to you during your section.

3 Imaging 1 Overview

The goal of this module is to use linear algebra techniques to capture real-world images with limited sensors. You will essentially be building a camera that captures one/multiple pixels at a time. While cameras in the real world are way more advanced than this, pixel imaging can be used in X-ray imaging and 3D imaging. It will also prove to be a useful aid to allow you to reinforce the linear algebra concepts you learn in class.

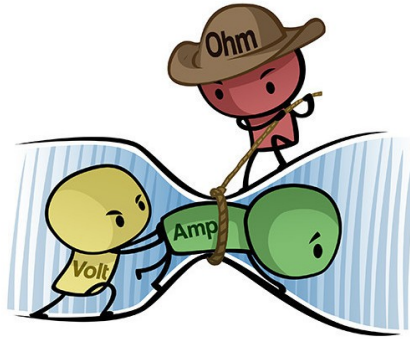
Here are some key terms/definitions that will help your understanding in lab next week:

- (a) If you try to walk across a carpet and then touch a metal doorknob, you might receive a small shock. This phenomenon of a "small shock" is commonly described as "static electricity". The shock is a result of **electric charge**. A basic property of matter, electric charge is a quantity of electricity that is caused by the balance (or imbalance) of electrons and protons in an atom.
Charge (Q) is measured in terms of its standard unit - **coulomb (C)**.
- (b) **Voltage (V) [volts (V)]** - Electric "pressure" that pushes charge through circuit.
- (c) **Current (I) [amperes (A)]** - Flow of charge through circuit.
1 Amp = 1 unit of charge per second = 1 coulomb per second
- (d) **Resistance (R) [ohms (Ω)]** - Resistance is a measure of how much opposition to the current flow a circuit component (such as a resistor) can provide (in Ohms).

The easiest way to understand Voltage, Current and Resistance is the **following analogy**:

Think of water flowing through a pipe. The voltage is the water force, the current is the amount of water flowing through the pipe, and the resistance is the size of the pipe.

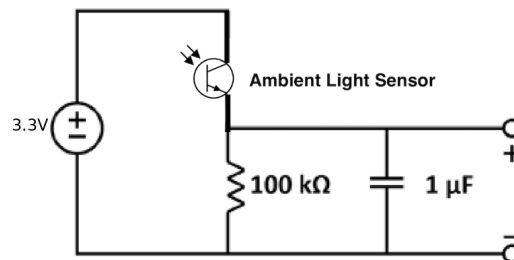
The more water will flow through the pipe (current), the more pressure is applied (voltage) and the bigger the pipe is (lower the resistance). There is a helpful visual aid at the top of the next page to better understand this relationship.



(e) Circuit Components:

- i. **Resistors** : circuit components that resist the flow of charge through a circuit.
- ii. **Capacitors**: circuit components that store your charge. They are called capacitors because they have a set capacity (in farads (F)). This set capacity, or, the value of a capacitor, is known as its **capacitance**. Capacitance is a measure of how much charge (energy) a capacitor can store.
- iii. **Voltage Source**: a device that can maintain a fixed voltage. Batteries are a simple example of a voltage source. Batteries are voltage sources that convert chemical energy into electrical energy to generate a voltage.
- iv. **Wires / Jumpers**. A wire is a piece of metal that is used to carry electric current in a circuit. Jumper cables are simply wires that are covered in insulation (a material that does not conduct electricity) and have a pin or a socket at each of its ends.
- v. **Light Emitting Diode (LED)**: A device that converts electric current directly into light.
- vi. **Ambient Light Sensor**: A device that detects the amount of light in its vicinity.

You will be building the following circuit in the lab:



NOTE: We don't expect you to understand anything about this circuit (yet). This image is simply a preview of what you will be building in the lab. Here are some values that are shown on the circuit diagram above:

$k\Omega$ = kiloOhm = 1000Ω

μF = microFarad = $10^{-6}F$

What does this circuit do? It is a circuit that reacts to light intensity. As you will see in the lab, it produces a clear, observable relationship between Voltage and Light Intensity, which you can use for your camera.

Contributors:

- Shreyash Iyengar
- Ayush Pancholy

- Anastasia Simonova
- Joyce Zhu
- Aniruddh Khanwale